



Physics Letters A

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Keywords

Real Hilbert space; Complex structure; Real commutant; Dual time; Inner-generative time; Positive energy; Zero modes; Operator algebra; Quantum kinematics

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The phase-selection architecture is summarized by the following reconstruction chain:

real orthogonal symmetry $\longrightarrow$ Real-commutant phase space,

inner-generative time flow $U(t)$

$= e^{tA} \longrightarrow$ dual energy orientation $\lambda > 0$,

selected phase $I \longrightarrow$ complex quantum information

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multipartite predictions by modifying or quotienting the composition rule rather than using the naive real tensor product [11], [12]. The present paper is orthogonal to that debate in scope. It does not defend the raw real tensor product as a complete many-body composition law. It reconstructs the compatible complex phase on a selected one-particle/operator-algebraic sector, and Section 4.1 states explicitly how the standard complex tensor product is recovered only after a J -balanced monoidal rule is imposed.

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The real commutant is

$$\text{Comm}_{\mathbb{R}}(U) = \{S \in B(H_{\mathbb{R}}) : SU(g) = U(g)S \text{ for all } g \in G\}. \quad (4)$$

Definition 2 Admissible phases

The admissible operators compatible with the structure are

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belongs to $\mathcal{A}_{\mathbb{R}}(U)$ exactly when

$$A_{tx}c_x = c_xA_x \tag{9}$$

for almost every x . Admissible phases are precisely the essentially bounded measurable fields satisfying (9) and

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1. The *inner-generative* or reversible role is the parameter t in the strongly continuous orthogonal group $U(t)$. Its real skew-adjoint generator is A , and its Stone-dual spectral parameter is λ , interpreted as energy or frequency.
2. Canonical conjugation exchanges the dual energies λ and $-\lambda$. A time orientation, equivalently the choice

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In the terminology of [Definition 3](#), A is the inner-generative time generator, $|A|$ is the positive rate or energy operator, and J_A is the orientation operator of the paired dual spectrum. The complex phase is therefore not an independent scalar postulate at this layer; it is the polar orientation of real inner-generative dynamics.

Corollary 1 Uniqueness in the Positive Factorization Class

Let $\tilde{J}$ be an orthogonal complex structure on H_{dom} and let $\tilde{B} \geq 0$ be self-adjoint with $\text{Dom}(\tilde{B}) = \text{Dom}(A) \cap H_{\text{dom}}$ such that

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For a larger group, for instance a semidirect product $K \rtimes \mathbb{R}$, positivity is not merely the sign choice in the time spectrum. The polar phase selected by the $\mathbb{R}$ -subgroup must also satisfy an equivariance condition with respect to the orientation-preserving part of the remaining symmetry. [Theorem 3](#) gives the basic criterion: symmetries preserving the oriented generator commute with J_A , while those reversing it implement the conjugate phase.

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selected, and it is exactly the usual scalar-balancing relation for the complex tensor product. It therefore recovers the underlying real Hilbert space of $(H_1, J_1) \widehat{\otimes}_{\mathbb{C}} (H_2, J_2)$ rather than independently deriving a new real multipartite theory. Recent formulations instead take an appropriate modified or symplectic composition rule as a central postulate for establishing operational equivalence with complex quantum mechanics. The constructions are compatible in purpose but occur at different logical stages.

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diagonal observer-accessible algebra and justify operational collapse only as an effective record approximation, in the usual spirit of decoherence and einselection [32]. This phase-before-records separation is part of the dependency discipline of the present framework: records may test, stabilize, or coarse-grain the selected J -linear algebra, but they do not supply an independent imaginary unit. Quantitative record stability is not proved here.

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This is the standard oscillator complex structure, but here it appears as the polar phase of the real time-evolution generator on the explicitly defined energy space.

5.2. Stationary Klein–Gordon fields

Let $\mathcal{h}$ be the real Hilbert space of spatial square-integrable fields on a static Cauchy surface, and let ω^2 be the positive self

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structure on the nonzero-energy sector by

$$A = J_A |A| = |A| J_A.$$

This selection does not require a gap at zero. Zero modes are precisely the sector on which the oriented time flow does not determine a phase.

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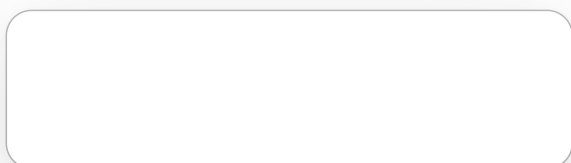
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Declaration of competing interest

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Completion principles, real commutants, and positive-frequency geometry for orthogonal Hilbert representations

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[18]

G.B. Folland

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